

Riccardo Rettore

Robotics & Physical AI engineer passionate about humanoids, embodied intelligence, and robot autonomy

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EXPERIENCE

- **Humanoid Robotics & Autonomous Systems Engineer @ EssilorLuxottica** May 2026 – Present
Quality Control Robotics Intern, Industrial Humanoid Deployment & Autonomous Test Automation Agordo, Italy
 - Researching and developing a **perception-to-action pipeline** on a **dual-arm ABB GoFa humanoid platform**, integrating **Vision-Language-Action (VLA) foundation models** with real-time closed-loop control for autonomous industrial manipulation.
 - Conducting research on large-scale **transformer-based world models (joint-embedding predictive architectures, action-conditioned video prediction)** as **zero-shot planners** for robot manipulation, benchmarked against **imitation learning** baselines (ACT, Diffusion Policy) trained via **LeRobot**.
 - Designing the **autonomous quality-control test architecture**, from **3D perception (FoundationPose, SAM2)** to closed-loop real-time verification in a live manufacturing environment.
- **Co-Founder & CTO @ Resit** Dec 2025 - May 2026
Smart Seating for healthier employees - resitchair.com Padova - San Francisco
 - Leading development of **learning-based posture classification models (LightGBM)**, transforming multi-modal sensor data (FSR, IMU, ToF) into real-time user state estimation and ergonomic insights.
 - Driving system architecture across **embedded sensing (ESP32/FreeRTOS)**, ML inference, and user-facing applications, enabling closed-loop intelligence from physical interaction to real-time feedback.
- **CDEI Group @ University of Catalunya** Feb 2025 – Aug 2025
Prof. A. Gracia (UPC), Prof. G. Susto (UniPd) Barcelona, Spain
 - Developed reinforcement learning-based navigation and control frameworks for autonomous robots using custom **Gymnasium** environments, implementing **PPO** and **SAC** in discrete and continuous action spaces with **Python** and **PyTorch**.
 - Integrated subsurface perception from Ground Penetrating Radar with surface sensing into decision-making pipelines, validating navigation policies in **ROS2**-based simulations with **Gazebo** and visualizing results in **RViz**.
- **Freelancer for Startups** Jan 2020 - Jan 2024
Full-Stack Developer Padova, Italy
 - Collaborated with startups to build cross-platform applications using **Flutter** and **Google technologies**.
 - Built both front-end interfaces and back-end systems to deliver seamless, user-friendly solutions.

EDUCATION

- **MSc in Control Systems Engineering** Oct 2023 - Oct 2025
Università degli Studi di Padova, Final Grade: 110/110 Padova, Italy
 - Final Thesis: [Leveraging Ground Penetrating Radar with Deep Reinforcement Learning for Subsurface-Informed Navigation in Autonomous Agricultural Robotics](#).
 - Relevant Coursework: Reinforcement Learning, Advanced Control Theory, Controls for Robotics Systems, Deep Learning, Machine Learning, Computer Vision, Autonomous Robotics.
 - Erasmus Exchange Program @ Universitat Politècnica de Catalunya, Barcelona, Spain: focused on Reinforcement Learning and Robotics, Physical AI and Autonomous Systems.
- **BSc in Mechatronics Engineering** Oct 2020 - July 2023
Università degli Studi di Padova, Final Grade: 109/110 Padova, Italy
 - Final Thesis: [Integration between ROS2 and Unity: performance analysis](#).
 - Relevant Coursework: Applied Mechanics, Analog Electronics, Electrical Machines and Drives, Instrumentation and Measurement, PLCs and Industrial Networks.

ACADEMIC EXPERIENCE

- **Aurora: Object Classification with Oriented Boosting**

July 2024 - Nov 2024

Neural Networks and Deep Learning project

Padova, Italy

- Built a 3D object classification pipeline with multi-task CNNs and oriented boosting, including voxelization, augmentation, labeling, and HDF5 preprocessing using **PyTorch**, **NumPy**, and **OpenCV**.
- Optimized GPU training and evaluated on ModelNet10/40, executing 3D data processing, CNN optimization, and computer vision pipelines.

- **Advanced Control Systems for Double-Wheel Robot**

April 2024 - June 2024

Control Engineering Laboratory Project

Padova, Italy

- Designed nominal and robust LQR controllers for a self-balancing two-wheeled robot, modeling dynamics via Lagrange formalism, linearizing around equilibrium, and implementing sensor fusion with complementary filtering in **MATLAB/Simulink**.
- Performed simulation and on-hardware testing of designed controllers, analyzing system stability, state-feedback, observer performance, and system robustness.

TECHNICAL SKILLS

- **Robotics & Simulation:** ROS2, Gazebo, RViz, MuJoCo, Isaac Gym / Isaac Lab, OpenCV
- **Robot Learning & Physical AI:** Imitation Learning (ACT, Diffusion Policy), Vision-Language-Action (VLA) models (OpenVLA, SmoVLA, π_0), World Models, Foundation Models, LeRobot, behavioral cloning, sim-to-real transfer, LIBERO benchmark
- **Perception:** FoundationPose, SAM2, CLIP, ResNet, 6D pose estimation, object segmentation, multi-modal sensor fusion
- **ML / RL Frameworks:** PyTorch, TensorFlow, Stable-Baselines3, Gymnasium, PPO, SAC, Transformers, LightGBM
- **Systems & Embedded:** Python, C++, RAPID, MATLAB/Simulink, ESP32/FreeRTOS, Bash/Linux, Git, Docker
- **Languages:** Italian (native), English (C1), Spanish (B1)